

FK-PINNs: Reproduction Audit

Noisy Feynman-Kac supervision helped the label story, but not the strongest quantitative claims.

Codex reproduction session[Ⓜ]

Trackio logbook · Hugging Face Jobs · paper #28407

Outcome. Claims 3 and 6 supported; Claims 1, 2, 4, and 5 not reproduced at tested scale.

0 Scope and Run Record

The audit used one-dimensional FK-representable Schrödinger, mean-exit-time, and committor PDEs. The local full run took 82 seconds; the completed HF GPU smoke run used a T4 and CUDA.

6
claims audited

3
toy PDEs

2
seeds/full

\$0.03
HF cost est.

STAGE 1
Euler-Maruyama FK labels

→

STAGE 2
PINN with uncertainty weights

1 Conditioning Grew

Fixed FK/data mass terms did not cap condition growth in the sine-operator audit, and the small neural Gauss-Newton proxy still grew/noisily worsened.

Audit	Slope	Verdict
Sine PINN	2.03	grows
Sine FK, lambda=1	2.03	grows
Neural PINN	0.99	grows
Neural FK	0.86	not bounded

Max. Sine FK condition: 1.04e6.

3 FK Label Noise Checks

Mean-exit-time label bias followed the expected square-root Euler trend and centered payoff noise was light-tailed.

Problem	Bias slope	Noise note
Mean exit	0.564	exp moment 9.81
Committer	0.331	bounded payoff

Supported. The $Y_{FK} = u^* + b + \text{zeta}$ decomposition was visible.

2 L2 Rate Not Visible

The scaled FK-PINN sweep was flat rather than decreasing over $N_{FK}=16..128$.

N=16	0.088 relative L2
N=32	0.106 relative L2
N=64	0.143 relative L2
N=128	0.091 relative L2

Fit. Slope +0.056, not negative.

4 Table 1 Factors Failed

Scaled 1D analogs did not match the reported improvement-factor pattern.

Problem	Paper	This audit
Schr.	6.5x	0.22x
MET	9.4x	0.31x
Committer	28.0x	1.46x

5 No Speedup Proxy

The GD proxy inherited condition growth.

N	PINN iters	FK iters
64	28k	28k
256	453k	448k
1024	7.24M	7.17M

No win. FK lambda=1 tracked PINN iteration growth.

6 Bottom Line

Yes	label noise, Algorithm 2
No	conditioning, rates, gains, speedup
Scope	scaled 1D audit

Full appendix-scale reproduction remains an open follow-up.

Read. Mechanism implemented; headline benefits were not robust.